



Keeping track of fresh fruit

A Bengaluru based startup has developed tech that uses AI to track the shelf life of produce. <https://digit.in/jun20-60>



IoT in healthcare

From disease management to tackling inefficiency, how IoT is helping problems in healthcare. <https://digit.in/jun20-61>



The catalysts of IoT

Commonly-used sensors in IoT devices

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Sensors have been around for a long time. The very first infrared sensors date back all the way to the 1940s while the thermostat was invented in the 1880s! Almost every industry and organisation today uses sensors to accomplish tasks. This prevalence of sensors saw a major push upon the advent of IoT (Internet of Things). IoT is quite a complex technology, with multiple architecture layers and a grid of connected devices that interact with each other to build functional applications. Most of these IoT devices or platforms rely on sensors to pass on data or intelligence to virtual dashboards. Recent advancement in IoT technology such as cellular IoT and LPWAN (Low-Power Wide Area Networking) are radically changing the way sensors

are used. From periodic measurements to real-time, continuous monitoring, sensors do it all in IoT devices. Sensors not only collect data, but also share it with a whole mesh of connected devices so that all of them can work together efficiently. They are the drivers behind your IoT devices functioning seamlessly and autonomously, and help your entire IoT system become 'smarter' over time. Now, let's learn about the various kinds of sensors commonly used in IoT devices.



TEMPERATURE SENSORS

Designed to measure the amount of heat energy in a source, temperature sensors are fine tuned to capture the slightest fluctuations in temperature and translate this information into readable data. These sensors are commonplace in the healthcare, manufacturing, and agriculture industries. While they were previously only used in A/C control, refrigerators and similar devices, after the

advent of IoT they have found a much wider range of uses. For example, temperature sensors are integrated in smart homes to assist homeowners to improve their energy consumption efficiency. IoT devices such as thermostats are fitted with these sensors and they get real-time temperature data from the house and accordingly adjust the temperature. In healthcare, temperature sensors enable no-contact temperature measurement. IoT-based temperature trackers measure the patient's body temperature and relay it to the physician upon asking or in a timely manner.



PROXIMITY SENSORS

Proximity sensors are common in IoT devices and are used in numerous applications. These sensors are capable of detecting the presence of nearby objects without physical contact. The way the proximity sensors are able to detect objects is by emitting an electromagnetic field or a beam of electromagnetic radiation which seeks any changes in